

The collective-unitary logarithmic-negativity rate of every bipartite spectrum

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Abstract

We determine the asymptotic maximum logarithmic negativity obtainable from tensor powers of an arbitrary bipartite density matrix by global unitaries on the original space. For fixed local dimensions $2 \leq m \leq n$ and spectrum p , the per-copy rate is the minimum over $1 \leq \alpha \leq 2$ of $[\log m + (\alpha - 1) \log n + \log \sum_i p_i^\alpha] / \alpha$. The result includes zero eigenvalues and unequal local dimensions. In the balanced case it reduces to one half of the logarithm of dimension times purity. We prove a one-shot spectral-prefix approximation using random Bell-basis projections and the matrix Bernstein inequality, then match a family of Schatten-norm upper bounds by spectral types and escort distributions. A fixed blockwise Bell output basis suffices at the exponential scale, with only the assignment of eigenvalues varying. An exact unequal-dimension example requires a strictly interior Rényi parameter. The rate is positive for every state except the maximally mixed state. This is a statement about logarithmic negativity under unrestricted collective global unitaries, not a claim about local distillation or efficient circuit implementation. Exact ancillary checks accompany the analytic proof; no Lean certification is asserted.

1. Statement, operation class, and relation to prior work

All logarithms are natural; rates in bits are obtained by dividing by $\log 2$.

Let $2 \leq m \leq n$ be fixed integers and let ρ be any density matrix on $\mathbb{C}^m \otimes \mathbb{C}^n$. Write $p = (p_1, \dots, p_{mn})$ for its eigenvalues, allowing zeros. For $k \geq 1$, put

$$F_k(\rho) = \max_{U \in \mathcal{U}((mn)^k)} \|(U \rho^{\otimes k} U^*)^{\Gamma_k}\|_1. \quad (1)$$

Here the tensor factors are grouped as $A^k : B^k$, of dimensions $m^k : n^k$, and Γ_k transposes the entire B^k factor in a product basis. The optimization allows arbitrary global unitaries on the existing space, but no additional ancilla, conditioning, measurement, or discarded subsystem. The maximum exists by compactness and continuity. Define the maximal logarithmic negativity by $E_k = \log F_k$, using the trace-norm convention of [VW], rather than the unlogged negativity $(F_k - 1)/2$.

Theorem 1 (exact rate). For every such state, including rank-deficient states, the limit exists and equals

$$\lim_{k \rightarrow \infty} \frac{E_k}{k} = \min_{1 \leq \alpha \leq 2} \frac{\log m + (\alpha - 1) \log n + \log \sum_i p_i^\alpha}{\alpha}. \quad (2)$$

The value at $\alpha = 1$ is $\log m$. In terms of the Renyi entropy $S_\alpha(p) = (1 - \alpha)^{-1} \log \sum_i p_i^\alpha$, the numerator for $\alpha > 1$ is $\log m + (\alpha - 1)(\log n - S_\alpha(p))$.

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Corollary 2 (balanced systems). If $m = n = d$, then

$$\lim_{k \rightarrow \infty} \frac{E_k}{k} = \frac{1}{2} \log(d^2 \text{Tr } \rho^2) = \log d - \frac{1}{2} S_2(\rho). \quad (3)$$

For every $2 \leq m \leq n$, the rate in (2) is zero exactly for the maximally mixed state; every other state has a positive rate.

Further conclusion (fixed-basis sufficiency). For each k , fix the blockwise Bell basis from (11), completed by product-coordinate vectors on the unused second-factor coordinates. Permuting the eigenvalues of $\rho^{\otimes k}$ among this fixed basis already attains the same rate (2). In the balanced case the entire output basis can be the standard generalized Bell basis. The proof below establishes this restricted lower bound directly; it does not assert exact finite-copy optimality of Bell-diagonal states.

This determines a regularized unitary-orbit entanglement measure, not a distillable-entanglement rate and not a local-operation protocol. It also does not give an exact one-copy maximum. Recent work [AMRS, v2] studies finite-dimensional negativity-from-spectrum bounds. Kondra et al. [K, Supplemental Lemma 6] already show finite-copy failure of absolute separability away from the maximally mixed state. Thus qualitative activation is not claimed as new here. The claim established below is the exact collective logarithmic-negativity exponent, including unequal local dimensions and every spectrum. Literature screening is not an exhaustive priority certification. No resolution of APPT=AS or of the unrestricted finite-copy APPT purity conjecture is asserted.

2. A finite-dimensional spectral envelope

For this section fix $2 \leq a \leq b$, write $N = ab$, and let $\lambda_1 \geq \dots \geq \lambda_N \geq 0$ sum to one. Define

$$L_r = \sum_{i=1}^r \lambda_i, \quad w_r = \min\{a, \sqrt{N/r}\}, \quad T_{a,b}(\lambda) = \max_{1 \leq r \leq N} w_r L_r. \quad (4)$$

Let $F_{a,b}(\lambda)$ be the maximum trace norm of the partial transpose along the corresponding global unitary orbit. Let $F_{a,b}^{\text{Bell}}(\lambda)$ restrict the output eigenbasis to the fixed basis just described, allowing only permutations of the eigenvalues. Also let $H_N = \sum_{r=1}^N 1/r$.

Theorem 3 (one-shot bounds). Every spectrum satisfies

$$\frac{T_{a,b}(\lambda)}{12 \log(8N)} \leq F_{a,b}^{\text{Bell}}(\lambda) \leq F_{a,b}(\lambda) \leq H_N T_{a,b}(\lambda). \quad (5)$$

In particular, the envelope approximates the logarithm of the orbit optimum up to $O(\log \log N)$, uniformly over spectra and aspect ratios. This does not assert a constant-factor approximation for the unlogged negativity.

2.1 Duality and the upper envelope

Partial transpose is an involution, preserves the Hilbert–Schmidt norm, and is self-adjoint for the trace pairing. For a Hermitian matrix X ,

$$\|X\|_1 = \max_{H=H^*, \|H\|_\infty \leq 1} \text{Tr}(HX). \quad (6)$$

If H is such a contraction, then

$$\|H^\Gamma\|_2 \leq \sqrt{N}, \quad \|H^\Gamma\|_\infty \leq a. \quad (7)$$

For the second inequality, take any unit vector v with Schmidt coefficients s_j . Its partially transposed rank-one projector has eigenvalues s_j^2 and $\pm s_i s_j$, so its trace norm is $(\sum_j s_j)^2 \leq a$. Consequently

$$|\langle v, H^\Gamma v \rangle| = |\text{Tr } H(|v\rangle\langle v|)^\Gamma| \leq a.$$

Taking the supremum over v proves (7). If $s_i(H^\Gamma)$ are the decreasing singular values, (7) implies $s_i(H^\Gamma) \leq w_i$. The singular-value trace inequality and (6) therefore give

$$F_{a,b}(\lambda) \leq \sum_i \lambda_i w_i \leq \sum_i \frac{L_i w_i}{i} \leq H_N T_{a,b}(\lambda). \quad (8)$$

The middle step uses the ordering, $i\lambda_i \leq L_i$.

2.2 A sharper Renyi family of upper bounds

For $1 < \alpha \leq 2$, put $q = \alpha/(\alpha - 1) \geq 2$. From (7),

$$\|H^\Gamma\|_q^q \leq a^{q-2} \|H^\Gamma\|_2^2 \leq a^{q-2} N.$$

Holder's inequality in (6) yields

$$F_{a,b}(\lambda) \leq \left(a b^{\alpha-1} \sum_i \lambda_i^\alpha \right)^{1/\alpha}. \quad (9)$$

At $\alpha = 1$ the same formula is the bound $F_{a,b} \leq a$. Thus (9) holds on the whole closed interval $[1, 2]$. There is no full-rank assumption.

3. Projections with a small partial-transpose norm

We prove the lower bound in (5) by constructing a projection of rank at least r . The rank need not equal r ; increasing the rank only increases the mass L_r used later.

Set $J = \lfloor b/a \rfloor$ and $N_0 = a^2 J$. Decompose the first aJ coordinate directions of the second factor into J disjoint blocks of dimension a . Let I_0 project onto $\mathbb{C}^a \otimes \mathbb{C}^{aJ}$. Since $b/a \geq 1$,

$$N/2 \leq N_0 \leq N. \quad (10)$$

No divisibility assumption on b is needed.

Let $\omega = \exp(2\pi i/a)$. For $0 \leq j < J$ and $0 \leq u, v < a$, define

$$|\phi_{juv}\rangle = \frac{1}{\sqrt{a}} \sum_{t=0}^{a-1} \omega^{ut} |t\rangle |ja + (t + v \bmod a)\rangle. \quad (11)$$

These N_0 vectors form an orthonormal basis of the range of I_0 . Indeed different blocks or shifts are orthogonal, and for the same block and shift the inner products are $a^{-1} \sum_t \omega^{(u'-u)t} = \delta_{u,u'}$.

Index them by i , and put $P_i = |\phi_i\rangle\langle\phi_i|$ and $W_i = P_i^\Gamma$. Each vector is maximally entangled inside its coordinate $a \times a$ block. The partial transpose of the standard Bell projector is the swap operator divided by a ; changing Bell vectors conjugates this matrix by a unitary. Hence, writing I_j for the identity on the corresponding product block,

$$\|W_i\|_\infty = 1/a, \quad W_i^2 = I_j/a^2, \quad \sum_i W_i = I_0, \quad \sum_i W_i^2 = I_0. \quad (12)$$

The last equality counts a^2 Bell vectors per block. The third follows from $\sum_i P_i = I_0$ and $I_0^\Gamma = I_0$. All statements hold for composite a as well as prime a .

Assume first $1 \leq r \leq N_0/8$. Choose independent Bernoulli variables ξ_i with parameter $p_0 = 4r/N_0 \leq 1/2$. The random matrix

$$P = \sum_i \xi_i P_i$$

is an orthogonal projection, because its rank-one summands are mutually orthogonal. Its rank $K = \sum_i \xi_i$ has expectation $4r$ and variance at most $4r$. Chebyshev's inequality gives

$$\Pr\{K < r\} \leq \frac{4}{9r} \leq \frac{4}{9}. \quad (13)$$

On the N_0 -dimensional range of I_0 , set

$$Y = P^\Gamma - p_0 I_0 = \sum_i X_i, \quad X_i = (\xi_i - p_0) W_i.$$

The summands are independent, centered, self-adjoint, satisfy $\|X_i\|_\infty \leq 1/a$, and by (12) have variance parameter

$$\left\| \sum_i \mathbb{E} X_i^2 \right\|_\infty = p_0(1 - p_0) \leq p_0. \quad (14)$$

Apply the self-adjoint matrix Bernstein bound [T, Theorem 1.4] to Y and $-Y$, then use the union bound:

$$\Pr\{\|Y\|_\infty \geq t\} \leq 2N_0 \exp\left(-\frac{t^2}{2(p_0 + t/(3a))}\right). \quad (15)$$

This is an application of a proved external theorem, not an inference from sampled matrices.

Put $L = \log(8N_0) > 1$ and $t = 2\sqrt{p_0 L} + 2L/a$. With $x = \sqrt{p_0 L}$ and $y = L/a$, the exact identity

$$t^2 - 2L(p_0 + t/(3a)) = 2x^2 + \frac{20}{3}xy + \frac{8}{3}y^2 \geq 0 \quad (16)$$

shows that the right side of (15) is at most $1/4$. Combining (13) and (15), the probability that either desired condition fails is at most $4/9 + 1/4 = 25/36 < 1$. There consequently exists a projection with $K \geq r$ and $\|P^\Gamma\|_\infty \leq p_0 + t$.

Since $p_0 \leq 1$ and $L \geq 1$,

$$p_0 + t \leq 3L\sqrt{p_0} + 2L/a \leq 12\log(8N) \max\{\sqrt{r/N}, 1/a\}. \quad (17)$$

For clarity, $\sqrt{p_0} = 2\sqrt{r/N_0} \leq 2\sqrt{2}\sqrt{r/N}$, and $6\sqrt{2} + 2 < 12$. Thus the constants in (17) are dimension independent.

4. Completing the finite-dimensional lower bound

For a rank- K projection supplied by Section 3, choose a global unitary so that $\sigma = U \text{diag}(\lambda) U^*$ has its K largest eigenvalues on the range of P . Choose the individual output eigenvectors to be the selected Bell vectors, the unselected Bell vectors, and the remaining product-coordinate vectors. Thus the output is diagonal in the same fixed basis for every input spectrum; only the permutation depends on the spectrum. This neither modifies the spectrum nor appends or discards a system.

Let $c = \|P^\Gamma\|_\infty > 0$. The matrix $H = P^\Gamma/c$ is a Hermitian contraction, so (6) and the trace-pairing identity give

$$\|\sigma^\Gamma\|_1 \geq \text{Tr}(\sigma^\Gamma P^\Gamma)/c = \text{Tr}(\sigma P)/c = L_K/c \geq L_r/c \geq \frac{w_r L_r}{12 \log(8N)}. \quad (18)$$

This proves the required lower bound for $r \leq N_0/8$.

If $r > N_0/8$, then (10) gives $w_r \leq \sqrt{N/r} < 4$, hence $w_r L_r \leq 4$. Every partial transpose of a density matrix has trace one, and therefore trace norm at least one. Thus (18), with its last lower bound, follows trivially in this range as well, since $12 \log(8N) > 4$. This also handles the cases where $N_0 < 8$ and no positive integer belongs to the small- r range. Taking the maximum over all r proves (5), including its fixed-basis restriction.

5. Tensor powers and spectral types

Let s denote the number of positive eigenvalues of ρ , and restrict distributions below to that support. For a probability vector q , write

$$H(q) = -\sum_i q_i \log q_i, \quad D(q\|p) = \sum_i q_i \log(q_i/p_i), \quad (19)$$

using $0 \log 0 = 0$.

Lemma 4 (type size). If kq_i are integers summing to k , the corresponding type has size

$$r_q = \frac{k!}{\prod_i (kq_i)!}, \quad (k+1)^{-s} e^{kH(q)} \leq r_q \leq e^{kH(q)}. \quad (20)$$

For a self-contained proof, consider multinomial sampling with probabilities q . The probability of its own type is $r_q e^{-kH(q)}$, which is at most one. This proves the upper bound. That type is a mode: for any other counts h_i , its probability divided by that of counts kq_i is

$$\prod_i \frac{(kq_i)!}{h_i!} (kq_i)^{h_i - kq_i}.$$

The factors $k^{kq_i - h_i}$ have canceled. Each displayed factor is at most one, by comparing the factorial factors with kq_i . If $q_i = 0$ but $h_i > 0$, the competing probability is zero and can be discarded; coordinates with both counts zero contribute one. Since there are at most $(k+1)^s$ types, the modal probability is at least $(k+1)^{-s}$. This proves the lower bound.

Every eigenvalue in type q equals $\exp(k \sum_i q_i \log p_i)$, so the total eigenvalue mass in that type is at least

$$(k+1)^{-s} e^{-kD(q\|p)}. \quad (21)$$

The sum of the largest r_q eigenvalues of $\rho^{\otimes k}$ is no smaller than the mass of this particular type. Applying (18) in dimensions $a = m^k, b = n^k$ and using the upper bound on r_q in (20), we obtain

$$F_k(\rho) \geq \frac{\exp\{kg(q)\}}{12 \log(8(mn)^k)(k+1)^s}, \quad g(q) = \min\left\{\log m, \frac{\log(mn) - H(q)}{2}\right\} - D(q\|p). \quad (22)$$

No assertion is made that a full type consists of exactly the largest eigenvalues. Only the elementary maximal-prefix comparison is used.

Every distribution q on the positive support is a limit of types $q^{(k)}$ with denominator k . Both H and $D(\cdot\|p)$ are continuous on this simplex. The logarithm of the denominator in (22), divided by k , tends to zero. Hence

$$\liminf_{k \rightarrow \infty} \frac{\log F_k}{k} \geq \max_q g(q). \quad (23)$$

On the other hand (9), applied to tensor powers, gives for every $1 \leq \alpha \leq 2$

$$\limsup_{k \rightarrow \infty} \frac{\log F_k}{k} \leq h(\alpha), \quad h(\alpha) = \frac{A + (\alpha - 1)B + \log Z(\alpha)}{\alpha}, \quad (24)$$

where $A = \log m$, $B = \log n$, and $Z(\alpha) = \sum_{p_i > 0} p_i^\alpha$.

6. Exact matching of the variational bounds

Let $\delta = B - A \geq 0$ and define the escort distribution

$$q_{\alpha,i} = p_i^\alpha / Z(\alpha), \quad 1 \leq \alpha \leq 2. \quad (25)$$

Direct differentiation on the positive support gives

$$h'(\alpha) = \frac{\delta - H(q_\alpha)}{\alpha^2}, \quad \frac{d}{d\alpha} H(q_\alpha) = -\alpha \operatorname{Var}_{q_\alpha}(\log p_i) \leq 0. \quad (26)$$

These formulas remain valid when some original eigenvalues are zero because all differentiation occurs on their fixed positive support.

If $H(p) \leq \delta$, then h is nondecreasing and its minimum is $h(1) = A$. Taking $q = p$ gives $g(p) = A$.

If $H(q_2) \geq \delta$, then h is nonincreasing and its minimum is $h(2)$. In this case

$$g(q_2) = \frac{A + B - H(q_2)}{2} - D(q_2\|p) = \frac{A + B + \log Z(2)}{2} = h(2). \quad (27)$$

In the remaining case $H(q_2) < \delta < H(p)$, continuity gives $\alpha_* \in (1, 2)$ with $H(q_{\alpha_*}) = \delta$. It minimizes h by (26). If the positive eigenvalues are not all equal, the variance in (26) is positive, so this parameter is unique. When all are equal, the strict intermediate case cannot occur. Write $\ell_\alpha = \sum_i q_{\alpha,i} \log p_i$. Since

$$H(q_\alpha) = -\alpha \ell_\alpha + \log Z(\alpha), \quad D(q_\alpha\|p) = (\alpha - 1)\ell_\alpha - \log Z(\alpha),$$

substitution at $H(q_{\alpha_*}) = \delta$ yields

$$g(q_{\alpha_*}) = A - D(q_{\alpha_*}\|p) = h(\alpha_*). \quad (28)$$

Boundary equalities are covered by either endpoint case. Equations (23)–(28) match the lower and upper bounds, prove existence of the limit, and prove Theorem 1. The lower bound in (22) already uses only fixed-basis eigenvalue permutations, so the same limit holds under that restriction. They also show directly that

$$\max_q g(q) = \min_{1 \leq \alpha \leq 2} h(\alpha). \quad (29)$$

No unproved exchange of minimization and maximization is used.

When $m = n = d$, $\delta = 0$ and $H(q_2) \geq 0$, so (27) proves Corollary 2. In general, if p is maximally mixed, every unitary fixes it, so $F_k = 1$. Conversely, if p is not uniform on all mn entries, then $H(p) < \log(mn)$; using $q = p$ in (23) gives the strictly positive lower bound

$$R \geq \min\{\log m, (\log(mn) - H(p))/2\} > 0. \quad (30)$$

7. Consequences and limits of the conclusion

For a pure input the rate is $\log m$, as it must be because a global unitary can map a pure vector to a maximally entangled vector. For a state uniform on a support of size r , (2) reduces to

$$R = \min\{\log m, \frac{1}{2} \log(mn/r)\}.$$

For general unequal dimensions an interior escort parameter may be essential; one cannot replace (2) merely by the minimum of its $\alpha = 1$ and $\alpha = 2$ endpoint values.

An exact interior example occurs on $\mathbb{C}^2 \otimes \mathbb{C}^8$. Put $v = 2^{4/3}$ and take

$$p = \frac{(v, 1, 1, 1, 1, 0, \dots, 0)}{v + 4}, \quad (31)$$

with eleven zero entries. At $\alpha = 3/2$, $v^{3/2} = 4$, so the escort law is $(1/2, 1/8, 1/8, 1/8, 1/8)$. Its entropy is $2 \log 2 = \log(8/2)$. Strict decrease in (26) proves that $\alpha_* = 3/2$ is the unique minimizer, and the exact rate is

$$R = \frac{11}{3} \log 2 - \log(4 + 2^{4/3}). \quad (32)$$

It is strictly smaller than both endpoint values. No numerical optimizer is used to establish this example.

In the balanced qutrit example with spectrum $(3, 1, 1, 1, 1, 1, 1, 1, 1)/11$, whose single-copy APPT property is proved in the preceding project, (3) gives

$$R = \frac{1}{2} \log(153/121) > 0.$$

Thus the orbit has zero one-copy logarithmic negativity but positive collective rate. The qutrit purity theorem is not used in the proof of Theorem 1; it only supplies a previously certified example. Rate measured with base-two logarithms is obtained by dividing every displayed rate by $\log 2$.

The theorem is existential over unrestricted global unitaries. It does not supply polynomial-size circuits or a local entanglement-distillation protocol. It does not prove that Haar-typical unitaries attain the optimum. The random selection in Section 3 is over a Bell basis and is used solely to prove existence. The result addresses all fixed local dimensions and all input spectra in the regularized problem; the exact finite-copy orbit maximum and the original higher-dimensional APPT purity conjecture remain separate problems.

References

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Verification scope and disclosures

The mathematical argument uses the stated external matrix Bernstein theorem, together with finite-dimensional matrix analysis. The ancillary program independently checks small cyclotomic Bell-projector identities, exact rational constants, type cardinalities, and algebraic identities; it also rejects deliberately corrupted inputs and source programs. These checks do not by themselves certify the unbounded analytic theorem. The present work is not Lean-formalized and has not been externally peer reviewed.

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The complete proof, source, and ancillary checks are available in the repository <https://github.com/mxym/math>, directory `research/collective-unitary-negativity-rate`. The preceding qutrit APPT purity theorem and its immutable Lean release are separate results and do not formally certify this paper.